In [75]:
```
import yaml
import matplotlib.pyplot as plt
from scipy.stats import skew, kurtosis, ks_2samp
from scipy.stats import wasserstein_distance as w1dist, pearsonr
from factor_model import load_model
from scenario_generator import *

with open("cfg.yaml") as f:
    _exp = yaml.safe_load(f)["experiment_name"]
PREFIX = f"model/{_exp}"
with open(f"{PREFIX}/cfg.yaml") as f:
    _cfg = yaml.safe_load(f)

NUM_GENERATE = _cfg["num_generate"]
rng = np.random.default_rng(42)
```

In [76]:
```
train_model = load_model(f"{PREFIX}")
test_model = load_model(f"{PREFIX}/test")
print(PREFIX)
```

```
model/DDPM
```

In [77]:
```
R_df = pd.read_parquet(train_model.data_source)
R_testdf = pd.read_parquet(test_model.data_source)
common_stocks = np.intersect1d(R_df['csecid'].unique(), R_testdf['csecid'].u
col_indices_test = np.searchsorted(np.sort(R_testdf['csecid'].unique()), com
col_indices = np.searchsorted(np.sort(R_df['csecid'].unique()), common_stock
R_testdf = R_testdf[R_testdf['csecid'].isin(common_stocks)]
R_df = R_df[R_df['csecid'].isin(common_stocks)]
R_train = R_df.pivot_table(index="date", columns="csecid", values="returns")
R_test = R_testdf.pivot_table(index="date", columns="csecid", values="returr
```

In [78]:
```
"""
R_df = pd.read_parquet(train_model.data_source)
assert np.array_equal(R_train.columns[col_indices], common_stocks), "Column
R_train = R_train.values
# test if the common stocks are correctly aligned
R_train = R_df.pivot_table(index="date", columns="csecid", values="returns")
R_test = R_testdf.pivot_table(index="date", columns="csecid", values="returr
assert R_test.columns.equals(R_train.columns), "Columns do not match common
"""
```

Out[78]:
```
'\nR_df = pd.read_parquet(train_model.data_source)\nassert np.array_equal(R
_train.columns[col_indices], common_stocks), "Column indices do not match c
ommon stocks"\nR_train = R_train.values\n# test if the common stocks are co
rrectly aligned\nR_train = R_df.pivot_table(index="date", columns="csecid",
values="returns")\nR_test = R_testdf.pivot_table(index="date", columns="cse
cid", values="returns")\nassert R_test.columns.equals(R_train.columns), "Co
lumns do not match common stocks"\n'
```

In [79]:
```
diffusionScenario = ScenarioGenerator(train_model, DiffusionSampler(f"{PREFI
R_gen = diffusionScenario.stock_generate(NUM_GENERATE)[:, col_indices]
R_gen.shape
```

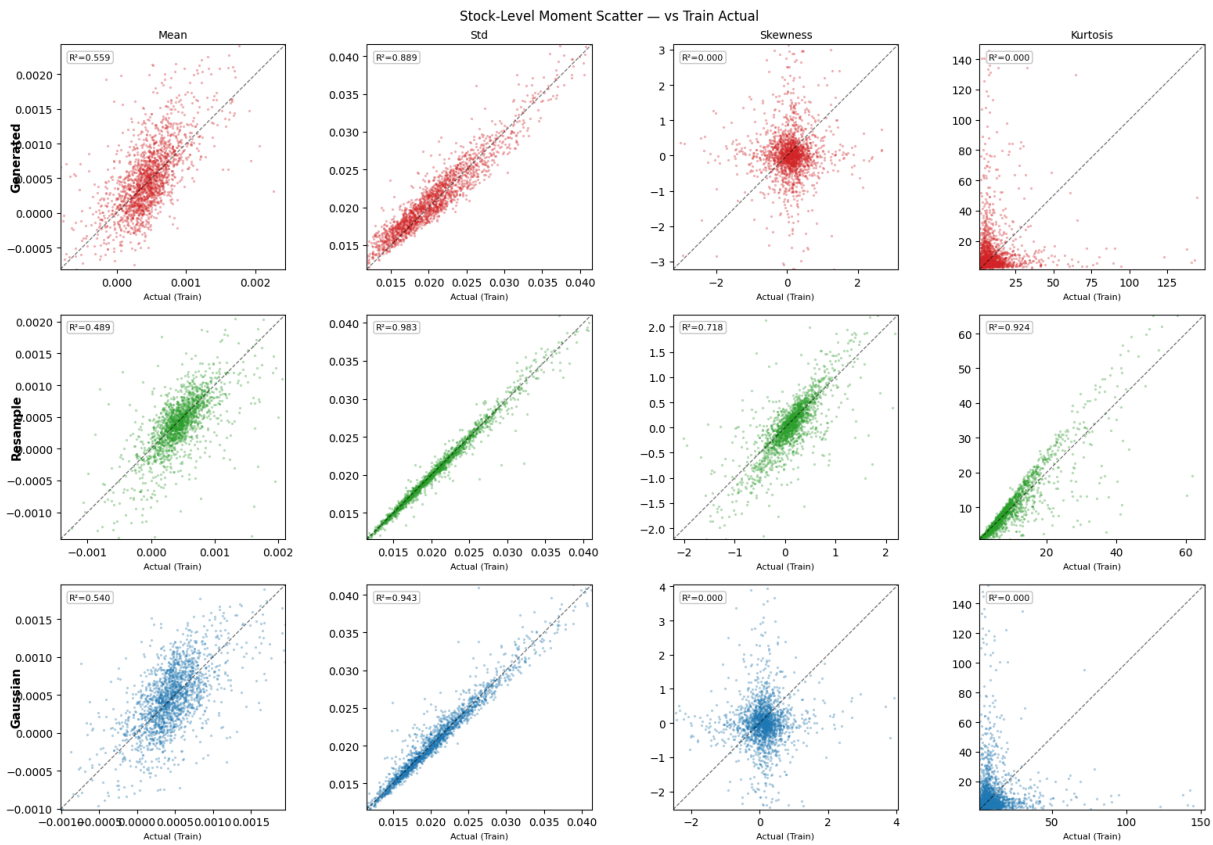
Out[79]: (4096, 1924)

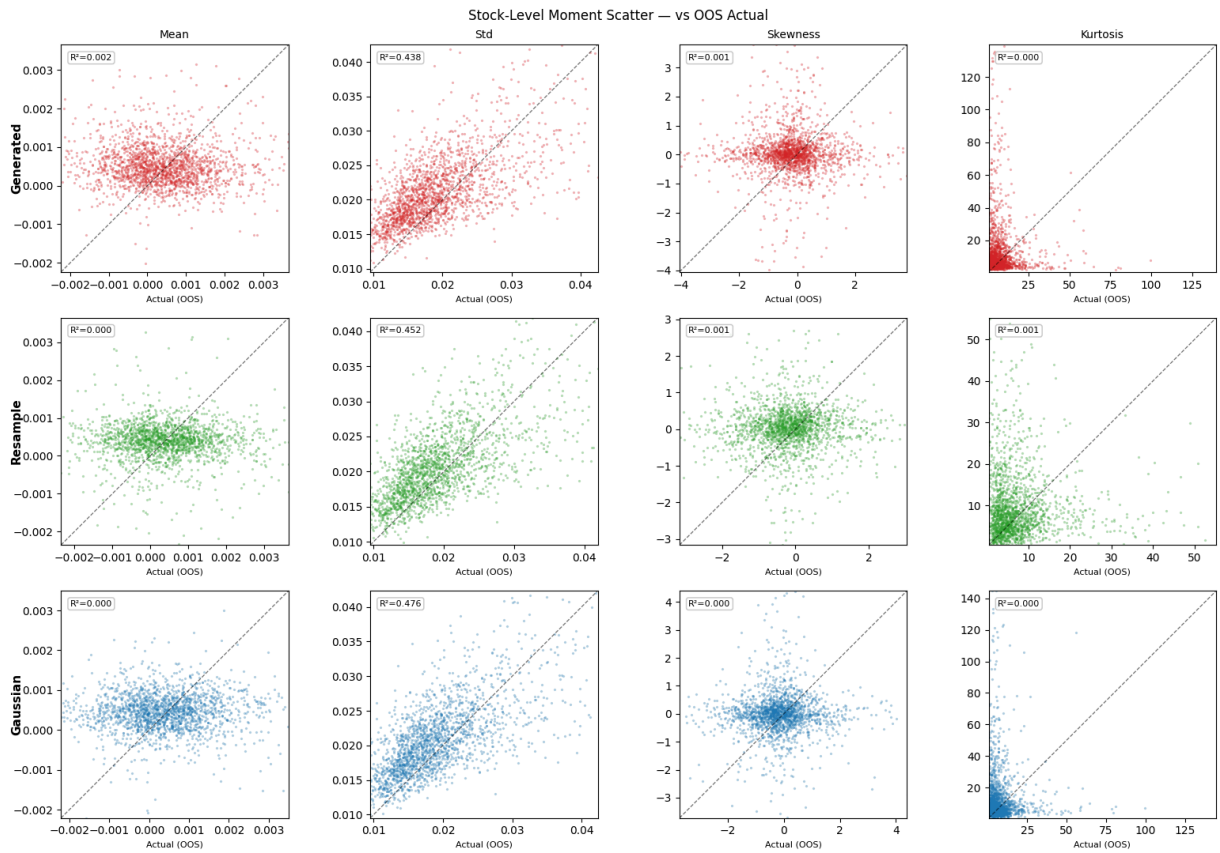
In [80]:
```
# get the baseline sample after filtering
row_idx = rng.choice(R_train.shape[0], size=NUM_GENERATE, replace=False)
R_rs = R_train[row_idx, :]
R_test = R_testdf.pivot_table(index="date", columns="csecid", values="returr
gaussionScenario = ScenarioGenerator(train_model, GaussianSampler(train_mode
R_gaussian = gaussionScenario.stock_generate(NUM_GENERATE)[:, col_indices]

print(f"Train returns: {R_train.shape}  NaN fraction: {np.isnan(R_train).mea
print(f"Test returns:  {R_test.shape}   NaN fraction: {np.isnan(R_test).mear
print(f"Resample returns: {R_rs.shape}  NaN fraction: {np.isnan(R_rs).mean()
print(f"Gaussion returns: {R_gaussian.shape} NaN fraction: {np.isnan(R_gauss
```

```
Train returns: (6262, 1924)  NaN fraction: 39.33%
Test returns:  (261, 1924)   NaN fraction: 4.27%
Resample returns: (4096, 1924)  NaN fraction: 39.35%
Gaussion returns: (4096, 1924) NaN fraction: 0.00%
```

In [81]:
```
## Stock-Level Moment Scatter Plots
MOMENTS = {
    "Mean":     lambda X: np.nanmean(X, axis=0),
    "Std":      lambda X: np.nanstd(X, axis=0),
    "Skewness": lambda X: skew(X, axis=0, nan_policy="omit"),
    "Kurtosis": lambda X: kurtosis(X, axis=0, nan_policy="omit"),
}

m_actual_tr  = {k: fn(R_train) for k, fn in MOMENTS.items()}
m_actual_oos = {k: fn(R_test)  for k, fn in MOMENTS.items()}
m_gen        = {k: fn(R_gen)       for k, fn in MOMENTS.items()}
m_rs         = {k: fn(R_rs)        for k, fn in MOMENTS.items()}
m_gs         = {k: fn(R_gaussian)  for k, fn in MOMENTS.items()}

MODELS = [
    ("Generated", m_gen, "#d62728"),
    ("Resample",  m_rs,  "#2ca02c"),
    ("Gaussian",  m_gs,  "#1f78b4"),
]

def _moment_scatter(actual_label, m_actual, fig_title):
    fig, axes = plt.subplots(3, 4, figsize=(16, 11))
    for row, (model_label, m_pred, color) in enumerate(MODELS):
        for col, (mname, _) in enumerate(MOMENTS.items()):
            ax = axes[row, col]
            x = m_actual[mname]
            y = m_pred[mname]
            mask = np.isfinite(x) & np.isfinite(y)

            lo = np.percentile(np.concatenate([x[mask], y[mask]]), 1)
            hi = np.percentile(np.concatenate([x[mask], y[mask]]), 99)

            ax.scatter(x[mask], y[mask], s=5, alpha=0.35, linewidths=0, colo
            ax.plot([lo, hi], [lo, hi], "k--", lw=0.9, alpha=0.55)

            r, _ = pearsonr(x[mask], y[mask])
            ax.text(0.04, 0.96, f"R²={r**2:.3f}",
```

```
                    transform=ax.transAxes, fontsize=8, va="top", ha="left",
                    bbox=dict(boxstyle="round,pad=0.25", facecolor="white",
                              edgecolor="0.75", alpha=0.85))

            ax.set_xlim(lo, hi)
            ax.set_ylim(lo, hi)
            ax.set_aspect("equal")
            ax.set_xlabel(f"Actual ({actual_label})", fontsize=8)
            if row == 0:
                ax.set_title(mname, fontsize=10)

        axes[row, 0].annotate(model_label, xy=(-0.22, 0.5), xycoords="axes f
                              fontsize=11, rotation=90, va="center", fontwei
    fig.suptitle(fig_title, fontsize=12)
    fig.tight_layout()
    plt.show()

_moment_scatter("Train", m_actual_tr,  "Stock-Level Moment Scatter — vs Trai
_moment_scatter("OOS",   m_actual_oos, "Stock-Level Moment Scatter — vs OOS
```





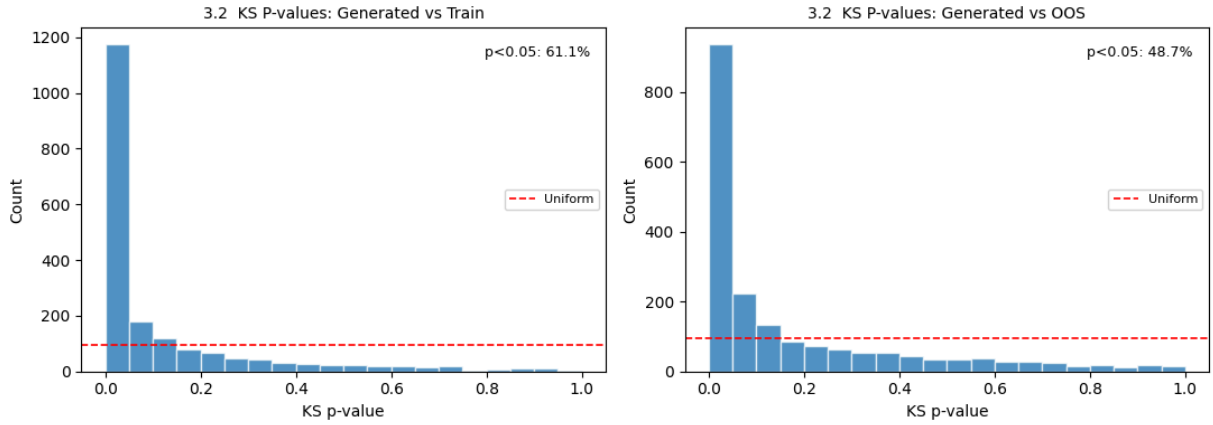
In [82]:

```
col_w = 14
header = f"{'':22}" + "".join(f"{m:<{col_w}}" for m in MOMENTS)
print("Pearson R²")
print(header)
print("-" * len(header))
for actual_label, m_actual in [("Train", m_actual_tr), ("OOS", m_actual_oos)
    for model_label, m_pred in [("Generated", m_gen), ("Resample", m_rs), ("
        label = f"{actual_label} {model_label}"
        row = f"{label:<22}"
        for mname in MOMENTS:
            x, y = m_actual[mname], m_pred[mname]
            mask = np.isfinite(x) & np.isfinite(y)
            r, _ = pearsonr(x[mask], y[mask])
            row += f"{r**2:<{col_w}.4f}"
        print(row)
    print()
```

```
Pearson R²
                       Mean          Std           Skewness      Kurtosis
-----------------------------------------------------------------------------
--
Train Generated        0.5585        0.8886        0.0000        0.0000
Train Resample         0.4895        0.9828        0.7184        0.9243
Train Gaussian         0.5404        0.9430        0.0000        0.0000

OOS Generated          0.0022        0.4377        0.0008        0.0000
OOS Resample           0.0000        0.4521        0.0011        0.0008
OOS Gaussian           0.0000        0.4756        0.0001        0.0005
```

In [83]:

```
## 3.2 Distribution of KS Test P-values
pvals_gen_train, pvals_gen_oos = [], []
pvals_rs_train,  pvals_rs_oos  = [], []
pvals_gs_train,  pvals_gs_oos  = [], []
S = R_train.shape[1]
for s in range(S):
    g   = R_gen[:, s];      g   = g[np.isfinite(g)]
    rs  = R_rs[:, s];       rs  = rs[np.isfinite(rs)]
    gs  = R_gaussian[:, s]; gs  = gs[np.isfinite(gs)]
    tr  = R_train[:, s];    tr  = tr[np.isfinite(tr)]
    oos = R_test[:, s];     oos = oos[np.isfinite(oos)]
    if len(g)  > 5 and len(tr)  > 5: pvals_gen_train.append(ks_2samp(g,  tr)
    if len(g)  > 5 and len(oos) > 5: pvals_gen_oos.append(  ks_2samp(g,  oos
    if len(rs) > 5 and len(tr)  > 5: pvals_rs_train.append( ks_2samp(rs, tr)
    if len(rs) > 5 and len(oos) > 5: pvals_rs_oos.append(   ks_2samp(rs, oos
    if len(gs) > 5 and len(tr)  > 5: pvals_gs_train.append( ks_2samp(gs, tr)
    if len(gs) > 5 and len(oos) > 5: pvals_gs_oos.append(   ks_2samp(gs, oos

pvals_gen_train = np.array(pvals_gen_train)
pvals_gen_oos   = np.array(pvals_gen_oos)
pvals_rs_train  = np.array(pvals_rs_train)
pvals_rs_oos    = np.array(pvals_rs_oos)
pvals_gs_train  = np.array(pvals_gs_train)
pvals_gs_oos    = np.array(pvals_gs_oos)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4))

for ax, pv, label in [(ax1, pvals_gen_train, "Generated vs Train"),
                      (ax2, pvals_gen_oos,   "Generated vs OOS")]:
    n_bins = 20
    ax.hist(pv, bins=n_bins, range=(0, 1), color="#1a6faf", alpha=0.75, edge
    ax.axhline(len(pv) / n_bins, color="red", linestyle="--", lw=1.2, label=
    ax.set_xlabel("KS p-value", fontsize=10)
    ax.set_ylabel("Count", fontsize=10)
    ax.set_title(f"3.2  KS P-values: {label}", fontsize=10)
    ax.legend(fontsize=8)
    frac = (pv < 0.05).mean()
    ax.text(0.97, 0.95, f"p<0.05: {frac:.1%}", transform=ax.transAxes,
            ha="right", va="top", fontsize=9)

fig.tight_layout()
plt.show()
```



In [84]:
```
col_w = 20
header = (f"{'':16}" + f"{'Generated':<{col_w}}"
          + f"{'Resample':<{col_w}}" + f"{'Gaussian':<{col_w}}")
print("KS p>0.05:")
print(header)
print("-" * len(header))
for label, pv_g, pv_r, pv_gs in [
    ("vs Train", pvals_gen_train, pvals_rs_train, pvals_gs_train),
    ("vs OOS",   pvals_gen_oos,   pvals_rs_oos,   pvals_gs_oos),
]:
    gen_val = f"{(pv_g  > 0.05).mean():.1%}"
    rs_val  = f"{(pv_r  > 0.05).mean():.1%}"
    gs_val  = f"{(pv_gs > 0.05).mean():.1%}"
    print(f"{label:<16}{gen_val:<{col_w}}{rs_val:<{col_w}}{gs_val:<{col_w}}"
```

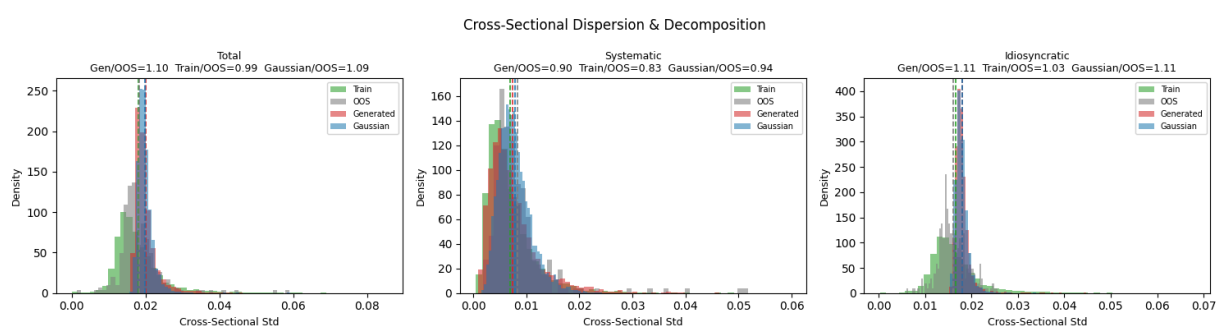
```
KS p>0.05:
                Generated           Resample            Gaussian
--------------------------------------------------------------------------
vs Train        38.9%               100.0%              29.8%
vs OOS          51.3%               71.2%               45.3%
```

In [85]:
```
beta_c    = train_model.beta[:, col_indices] # beta
res_std_c = train_model.res_std[col_indices]

# training part construction
sys_train  = train_model.F.values.astype(np.float32) @ beta_c
idio_train = train_model.residuals[:, col_indices].astype(np.float32)
# test part construction
test_beta_c = test_model.beta[:, col_indices_test]
sys_oos  = test_model.F.values.astype(np.float32) @ test_beta_c
idio_oos = test_model.residuals[:, col_indices_test].astype(np.float32)
```

In [86]:
```
assert np.allclose(sys_train + idio_train, R_train[1:, :], equal_nan=True)
assert np.allclose(sys_oos + idio_oos, R_test[1:, :], equal_nan=True)
```

In [87]:
```
# helper: factor samples -> (sys, idio) for common stocks
def _decompose(sampler, num, model, beta_c, res_std_c, seed):
    fs_raw = sampler.generate(num).astype(np.float32)
    if model.factor_type == "regression":
        fs_full = np.column_stack([np.ones((len(fs_raw), 1)), fs_raw]).astyp
    else:
        fs_full = fs_raw
```

```
    sys = fs_full @ beta_c
    t_noise = np.random.default_rng(seed).standard_t(model.res_df, size=sys.
    idio = t_noise * np.sqrt((model.res_df - 2) / model.res_df) * res_std_c
    return sys, idio

sys_gen,      idio_gen      = _decompose(diffusionScenario.sampler, NUM_GENE
sys_gaussian, idio_gaussian = _decompose(gaussionScenario.sampler,  NUM_GENE

panels = [
    ("Total",         R_gen,    R_test,  R_train),
    ("Systematic",    sys_gen,  sys_oos, sys_train),
    ("Idiosyncratic", idio_gen, idio_oos, idio_train),
]

gs_mats = {
    "Total":         R_gaussian,
    "Systematic":    sys_gaussian,
    "Idiosyncratic": idio_gaussian,
}

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
for ax, (comp, gmat, omat, tmat) in zip(axes, panels):
    gsmat = gs_mats[comp]
    gv  = np.nanstd(gmat,  axis=1)
    gsv = np.nanstd(gsmat, axis=1)
    ov  = np.nanstd(omat,  axis=1)
    tv  = np.nanstd(tmat,  axis=1)
    ax.hist(tv,  bins=50, density=True, alpha=0.55, color="#2ca02c", label="
    ax.hist(ov,  bins=50, density=True, alpha=0.55, color="#7f7f7f", label="
    ax.hist(gv,  bins=50, density=True, alpha=0.55, color="#d62728", label="
    ax.hist(gsv, bins=50, density=True, alpha=0.55, color="#1f78b4", label="
    for val, col in [(tv.mean(), "#2ca02c"), (ov.mean(), "#7f7f7f"),
                     (gv.mean(), "#d62728"), (gsv.mean(), "#1f78b4")]:
        ax.axvline(val, color=col, linestyle="--", lw=1.2)
    r_gen = gv.mean()  / (ov.mean() + 1e-12)
    r_tr  = tv.mean()  / (ov.mean() + 1e-12)
    r_gs  = gsv.mean() / (ov.mean() + 1e-12)
    ax.set_xlabel("Cross-Sectional Std", fontsize=9)
    ax.set_ylabel("Density", fontsize=9)
    ax.set_title(comp + "\nGen/OOS=%.2f  Train/OOS=%.2f  Gaussian/OOS=%.2f"
    ax.legend(fontsize=7)

fig.suptitle("Cross-Sectional Dispersion & Decomposition", fontsize=12)
fig.tight_layout()
plt.show()
```



In [88]:
```
row_idx_r = row_idx[row_idx < len(sys_train)]
sys_rs    = sys_train[row_idx_r]
idio_rs   = idio_train[row_idx_r]

rs_panels = [
    ("Total",         R_rs,    R_test),
    ("Systematic",    sys_rs,  sys_oos),
    ("Idiosyncratic", idio_rs, idio_oos),
]

col_w = 20
header = (f"{'Component':<16}{'W1(Gen, OOS)':<{col_w}}"
          + f"{'W1(Gaussian, OOS)':<{col_w}}{'W1(Resample, OOS)':<{col_w}}")
print("Wasserstein-1 Distance of Cross-Sectional Std (vs OOS):")
print(header)
print("-" * len(header))
for (comp, gmat, omat, _), (_, rsmat, _) in zip(panels, rs_panels):
    gsmat = gs_mats[comp]
    gv  = np.nanstd(gmat,  axis=1); gv  = gv[np.isfinite(gv)]
    gsv = np.nanstd(gsmat, axis=1); gsv = gsv[np.isfinite(gsv)]
    ov  = np.nanstd(omat,  axis=1); ov  = ov[np.isfinite(ov)]
    rsv = np.nanstd(rsmat, axis=1); rsv = rsv[np.isfinite(rsv)]
    print(f"{comp:<16}{w1dist(gv, ov):<{col_w}.4f}{w1dist(gsv, ov):<{col_w}.
```

```
Wasserstein-1 Distance of Cross-Sectional Std (vs OOS):
Component       W1(Gen, OOS)        W1(Gaussian, OOS)   W1(Resample, OOS)
--------------------------------------------------------------------------
Total           0.0023              0.0029              0.0013
Systematic      0.0010              0.0010              0.0014
Idiosyncratic   0.0022              0.0022              0.0014
```